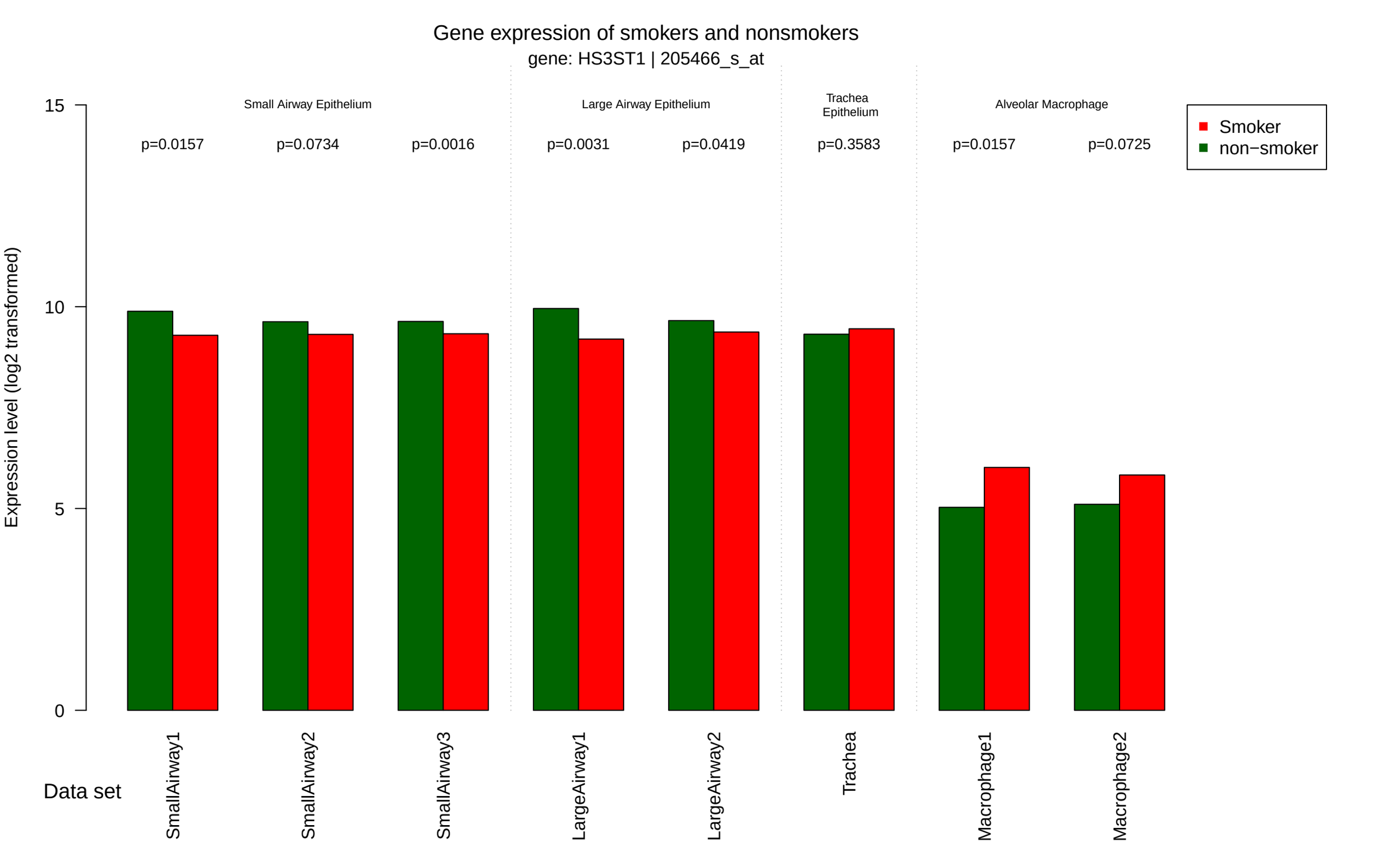
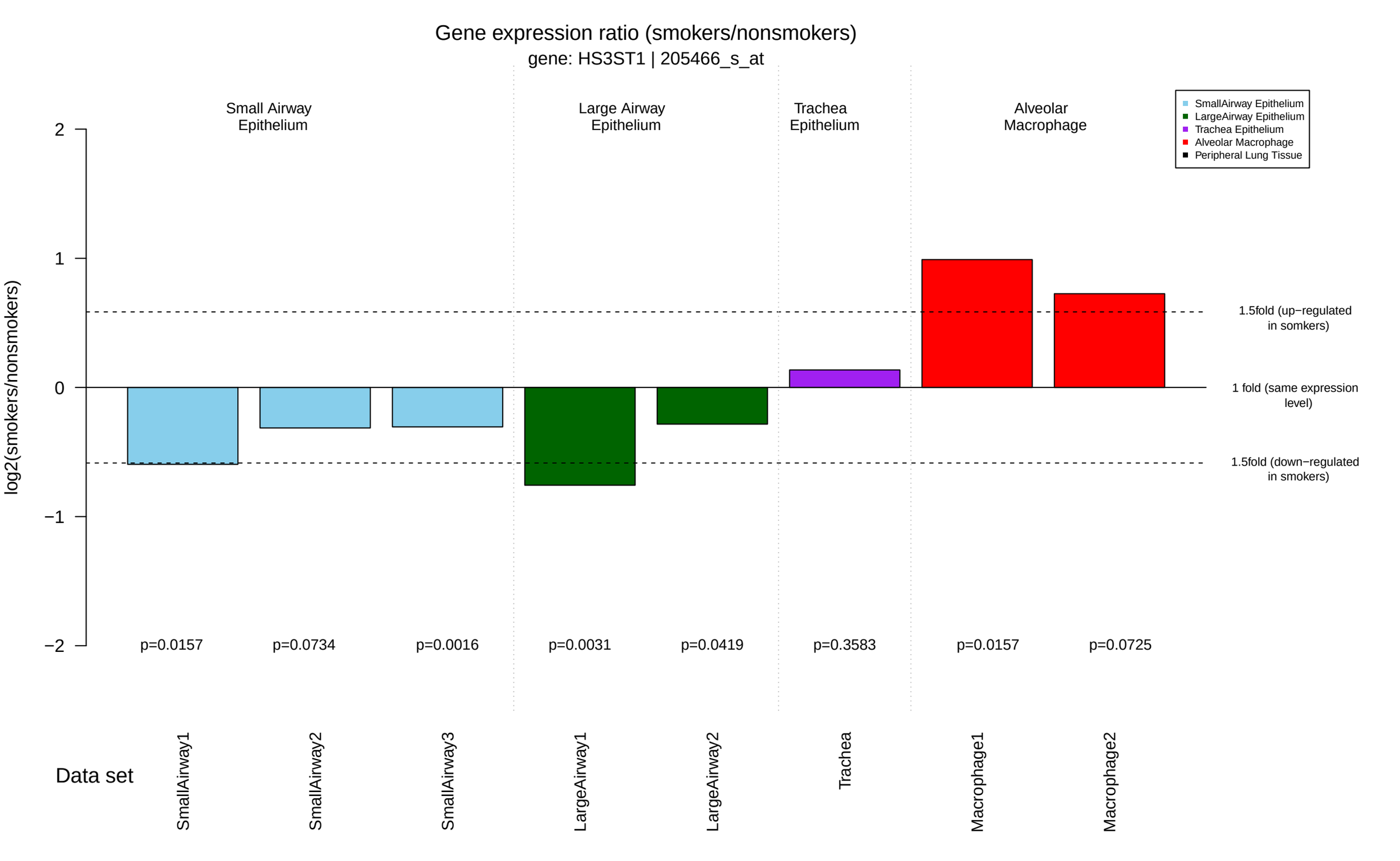
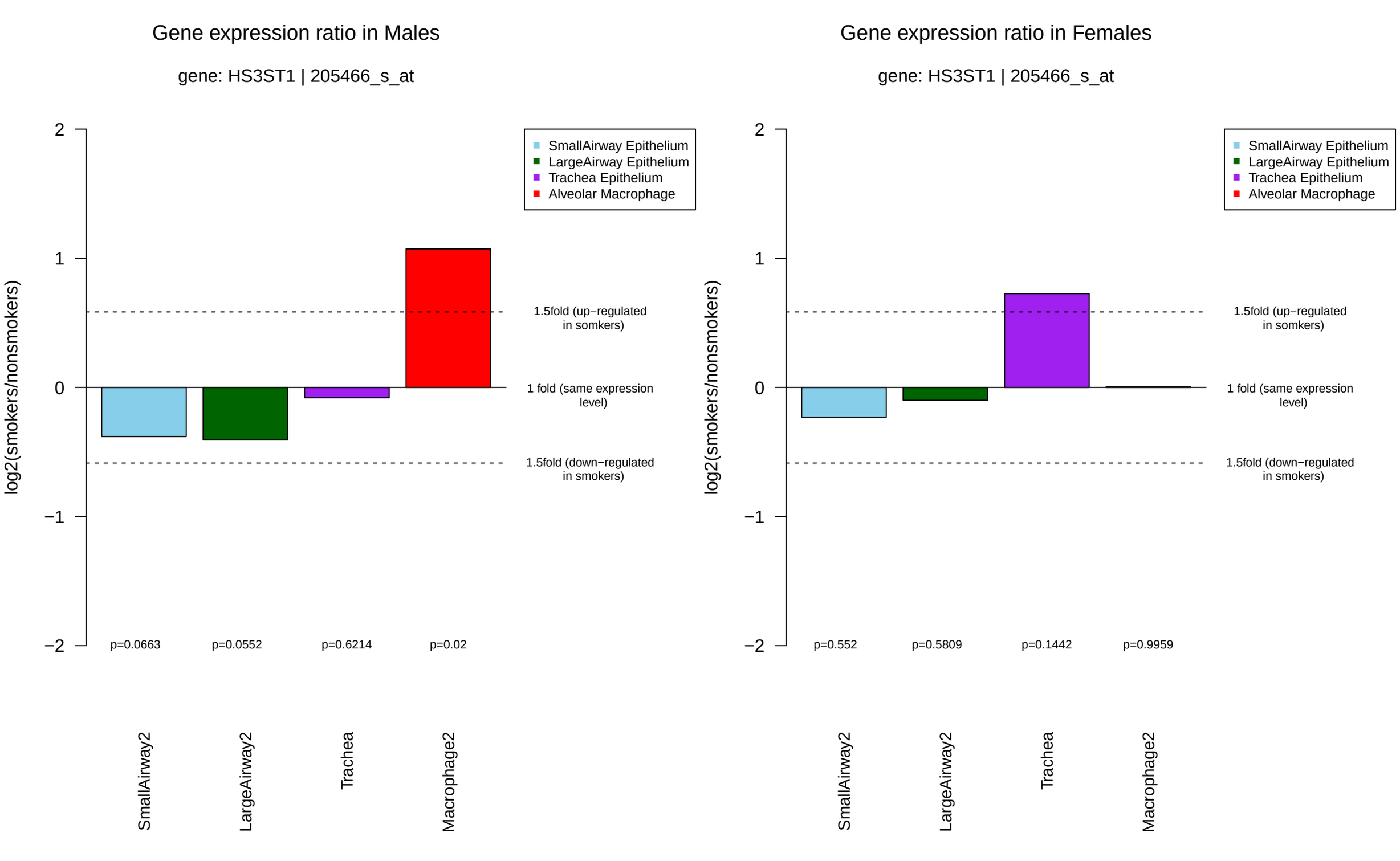
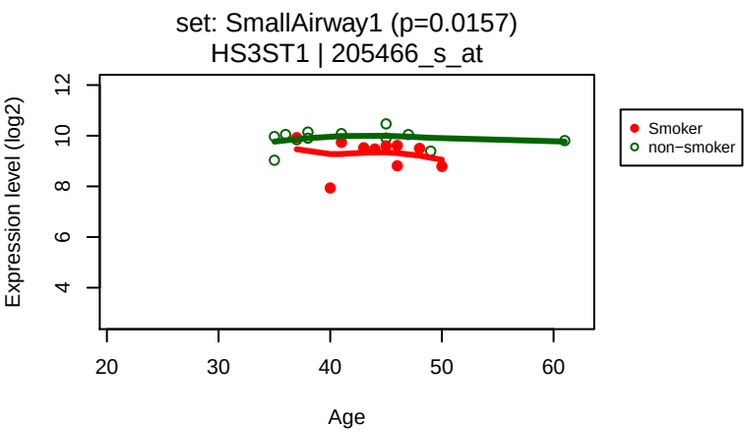
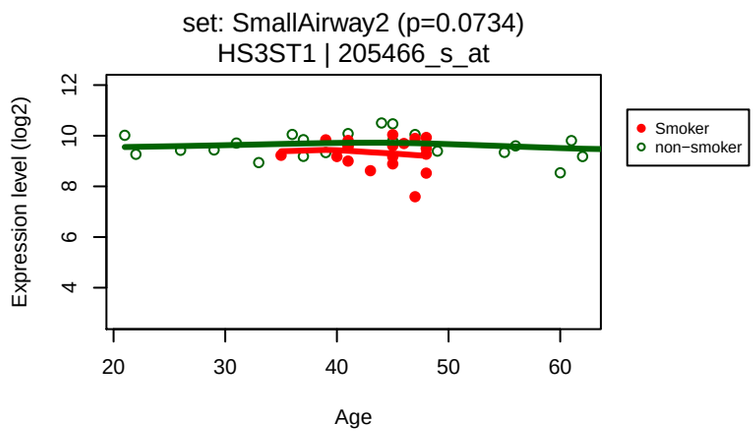
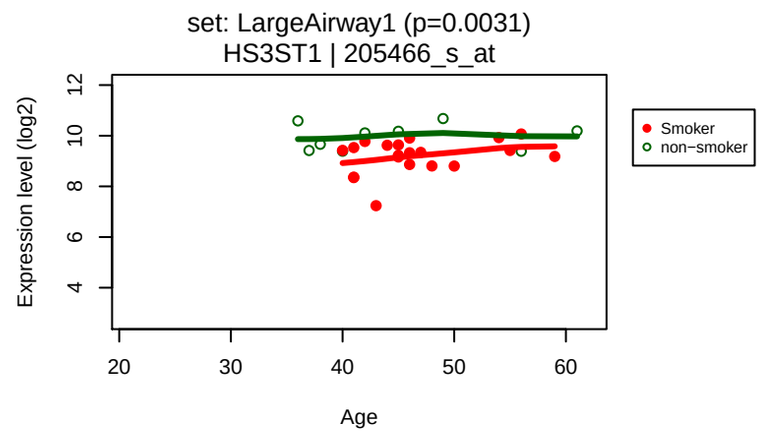

Gene expression of smokers and nonsmokers
gene: HS3ST1 | 205466_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0157
p=0.0734
p=0.0016
p=0.0031
p=0.0419
p=0.3583
p=0.0157
p=0.0725
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

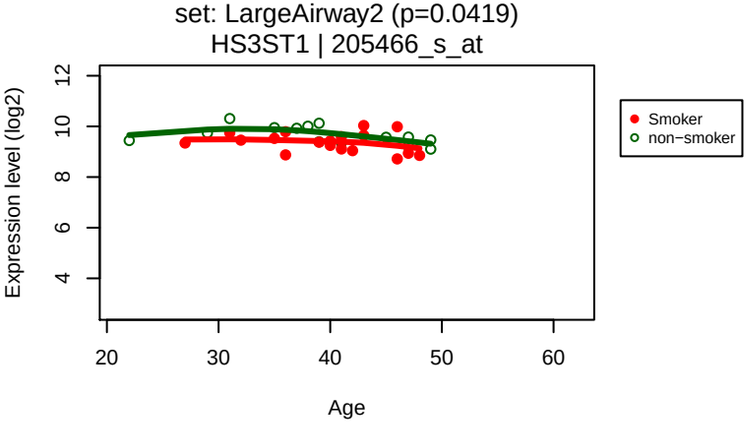

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST1 | 205466_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0157
p=0.0734
p=0.0016
p=0.0031
p=0.0419
p=0.3583
p=0.0157
p=0.0725
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

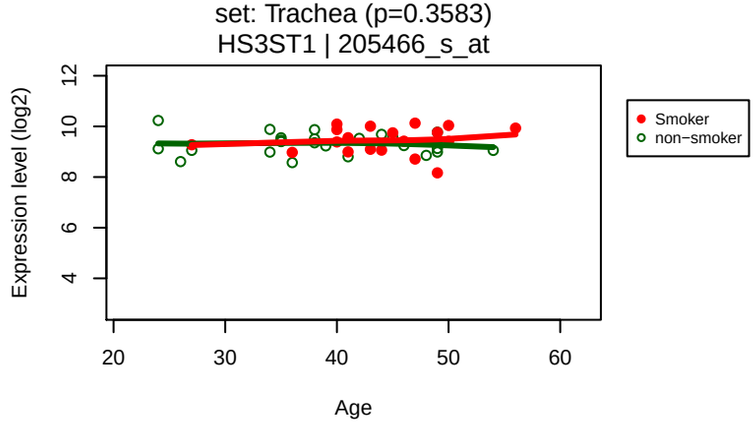

Gene expression ratio in Males
gene: HS3ST1 | 205466_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0663
p=0.0552
p=0.6214
p=0.02
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST1 | 205466_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.552
p=0.5809
p=0.1442
p=0.9959
SmallAirway2
LargeAirway2
Trachea
Macrophage2

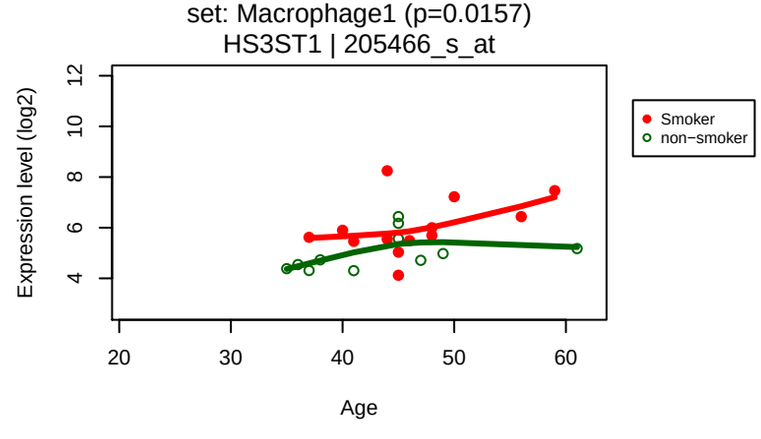

set: SmallAirway1 (p=0.0157)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker

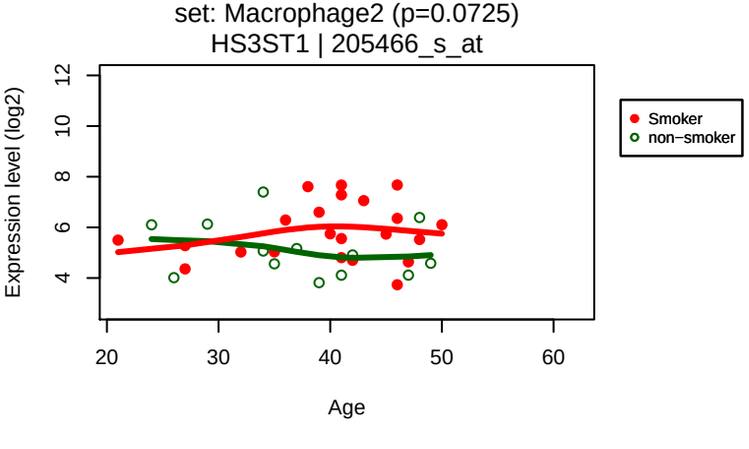

set: SmallAirway2 (p=0.0734)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker

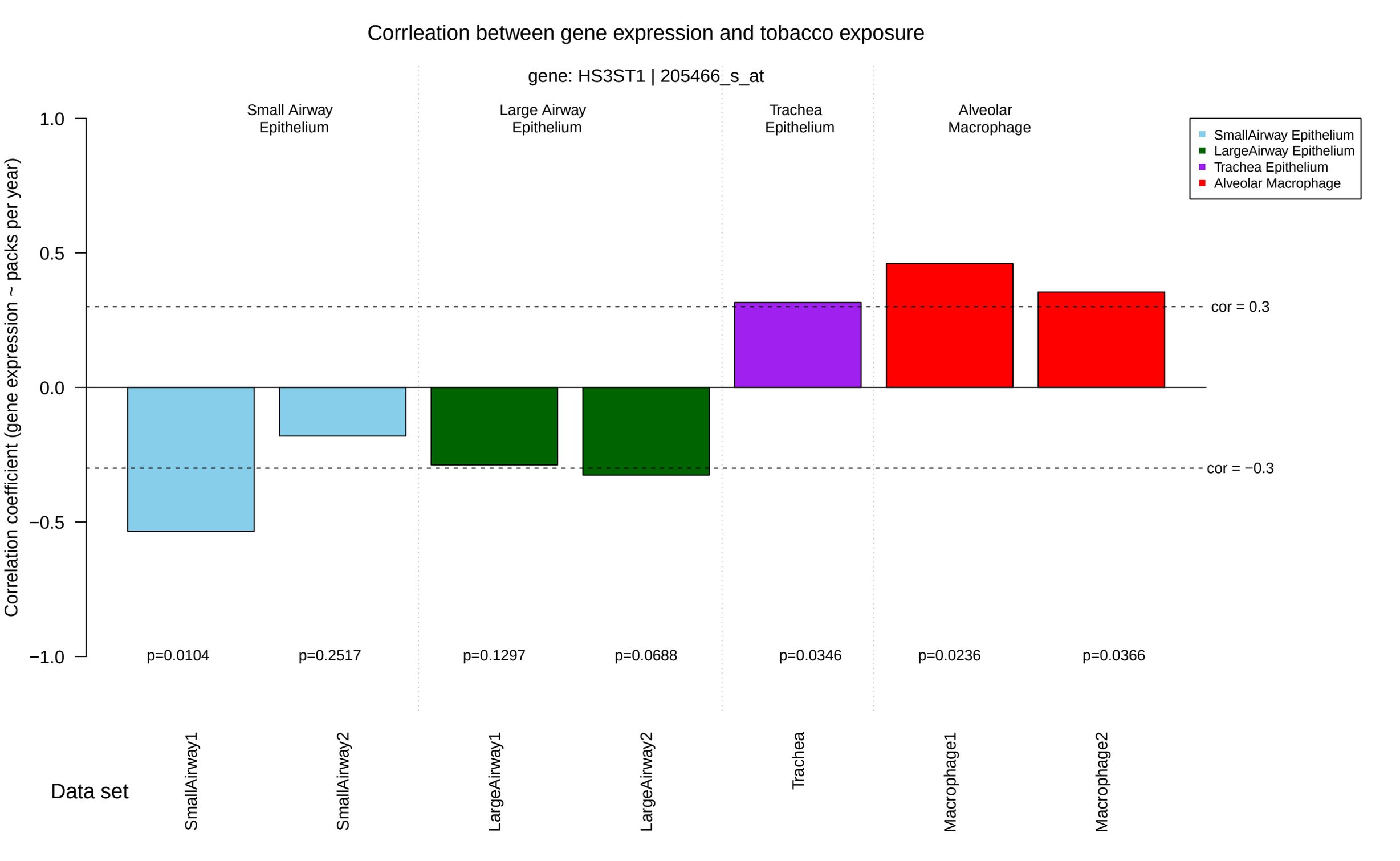

set: LargeAirway1 (p=0.0031)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0419)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3583)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0157)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0725)
HS3ST1 | 205466_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST1 | 205466_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0104
p=0.2517
p=0.1297
p=0.0688
p=0.0346
p=0.0236
p=0.0366
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2